Samuele Ferracin

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Google Scholar | LinkedIn | Portfolio

EDUCATION

University of Pennsylvania Philadelphia, PA | Sept 2021 - Present

Ph.D. in Mechanical Engineering

- Research focus: Nonlinear dynamics of metamaterials, nonlinear FEA (Abaqus), multistable structures, fluid-structure interaction, DIW 3D printing

Polytechnic University of Turin Turin, Italy | 2018 - 2021

M.Sc. in Aerospace Engineering - Propulsion Systems

- GPA: 106/110 (~3.85/4.0), Top 5%, merit-based scholarship recipient
- Coursework: structural mechanics, propulsion systems, supersonic flow, CFD of turbomachinery

Polytechnic University of Turin Turin, Italy | 2015 - 2018

B.Sc. in Aerospace Engineering

- GPA: 106/110 (~3.85/4.0), Top 3%
- Honors: “Young Talents” Program (top ~4% admitted) + scholarship
- Coursework: Mathematics, physics, structural mechanics, thermodynamics, aerodynamics, materials science, flight mechanics, and aerospace structures. Advanced electives: quantum physics, complex analysis, electromagnetic physics

RESEARCH and TEACHING EXPERIENCE

University of Pennsylvania, Mechanical Engineering and Applied Mechanics Philadelphia, PA | Sept 2021 - Present

Ph.D. Student Researcher (Advisor: Prof. Jordan Raney)

- Developed and validated nonlinear Abaqus FEA models (large deformation, buckling/post-buckling, contact, friction, dynamic response) for multistable structures
- Built coupled fluid-structure interaction simulations to predict structural behavior and flow-induced deformation of multistable metamaterials in unsteady flow for drag reduction (DoD-funded MURI), validated with wind tunnel experiments
- Designed, built, and optimized novel multistable metamaterial using parametric FEA, enabling contact-based tunable stiffness
- Designed and experimentally validated in wind tunnel bistable vortex generators and multistable hinges for flow separation control and wing surface morphing
- Developing data-driven technique using structure-preserving neural networks to improve fluid-structure interaction models
- Developed micro-scale metamaterials, modeling stability, deformation and use for precise compression of biological cells
- Conceptualized and numerically studied a new control strategy for transition waves in bistable metamaterials for scalable mechanical memories
- Developing 3D printed soft artificial contractile muscles with conductive elastomer composites for robotics and prosthetics

Stony Brook University Stony Brook, NY | Nov 2020 - Jul 2021

M.Sc. Thesis (Advisor: Prof. Paolo Celli)

- Designed, experimentally tested and simulated (Abaqus) the buckling/post-buckling dynamics of a bistable vibration induced deployable morphing structure for space applications
- Supported by a merit-based research fellowship

Polytechnic University of Turin, Superconductivity Lab Turin, Italy | 2018 - 2020

Junior Research Fellow (Advisor: Prof. Francesco Laviano)

- Designed, tested and simulated (COMSOL) thermo-electromagnetic response of a superconducting bolometer
- Conceptualized and modeled a superconducting magnetic energy storage system (SMES) with ENI S.p.A.

PROFESSIONAL EXPERIENCE

Ladar Ltd. Oslo, Norway | Aug - Sept 2018

Summer Engineering Intern

- Designed (Autodesk Inventor) a laser alignment mechanism for maritime LIDAR sensors

Tielogic S.r.l. Solagna, Italy | Jul - Sept 2016

Summer Engineering Intern

- Developed custom optical inspection tools for quality control
- Rapid prototyping with FDM 3D printing

SELECTED PUBLICATIONS

1. **S. Ferracin**, D. Jin, V. Tournat, J.R. Raney. *Phonon controlled mechanical memory via pinning and depinning of transition waves.* Under review at Physical Review Letters, 2026.
2. M.D. Bathgate, P.J. Ansell, **S. Ferracin**, D. Jin, S. Patel, J.R. Raney. *Airfoil Decambering for Gust Load Alleviation Using a Bi-stable Hinge.* AIAA SCITECH, 2026.
3. D. Jin, **S. Ferracin**, V. Tournat, S. Patel, P.K. Purohit, J.R. Raney. *Transition waves in a beam coupled to a foundation of symmetric bistable elements*. Physical Review Applied, 2025.
4. A.B.M.T. Haque*, **S. Ferracin***, J.R. Raney. *Reprogrammable mechanics via individually switchable bistable unit cells in a prestrained chiral metamaterial*. Advanced Materials Technologies, 2024.
5. A. Rahman, **S. Ferracin**, S. Tank, C. Zhang, P. Celli. *Shape-retaining beam-like morphing structures via localized snap through*. International Journal of Solids and Structures, 2024.
6. Q. He*, **S. Ferracin***, J.R. Raney. *Programmable responsive metamaterials for mechanical computing and robotics*. Nature Computational Science, 2024.

Full list on Google Scholar *First co-authors

CONFERENCES

- **THz Sapienza 2019** - Oral presentation
- **SES 2023**, Minneapolis, MN - Oral presentation
- **SES 2025**, Atlanta, GA - Oral presentation
- **FMI Workshop 2026,** Dayton, OH - Oral presentation and poster
- **USNC-TAM 2026**, Pasadena, CA - Oral presentation - Recipient of American Academy of Mechanics Travel Fellowship

TEACHING and PEER EXPERIENCE

- MEAM 5020 - Energy Engineering in Power Plants and Transportation Systems (2022)
- MEAM 3210 - Dynamic Systems and Control (2023, 2024, 2026)
- Led lectures, recitations, and grading for graduate and undergraduate mechanical engineering students
- Peer reviewer for PRX

TECHNICAL SKILLS

- **FEA**: Abaqus, COMSOL (nonlinear analysis, large deformations, contact, friction, dynamic simulations, buckling/post-buckling, explicit and implicit, parametric analysis, experimental validation)
- **Multiphysics**: Fluid-structure interaction, thermo-electromagnetic simulations
- **CAD**: SolidWorks, Inventor
- **Prototyping**: 3D printing (FDM, SLA, DIW)
- **Programming**: MATLAB, Python
- **Languages**: Italian (native), English (fluent), Spanish (basic)